

ARTIFICIAL INTELLIGENCE

Lecture Set 21
Q-Learning

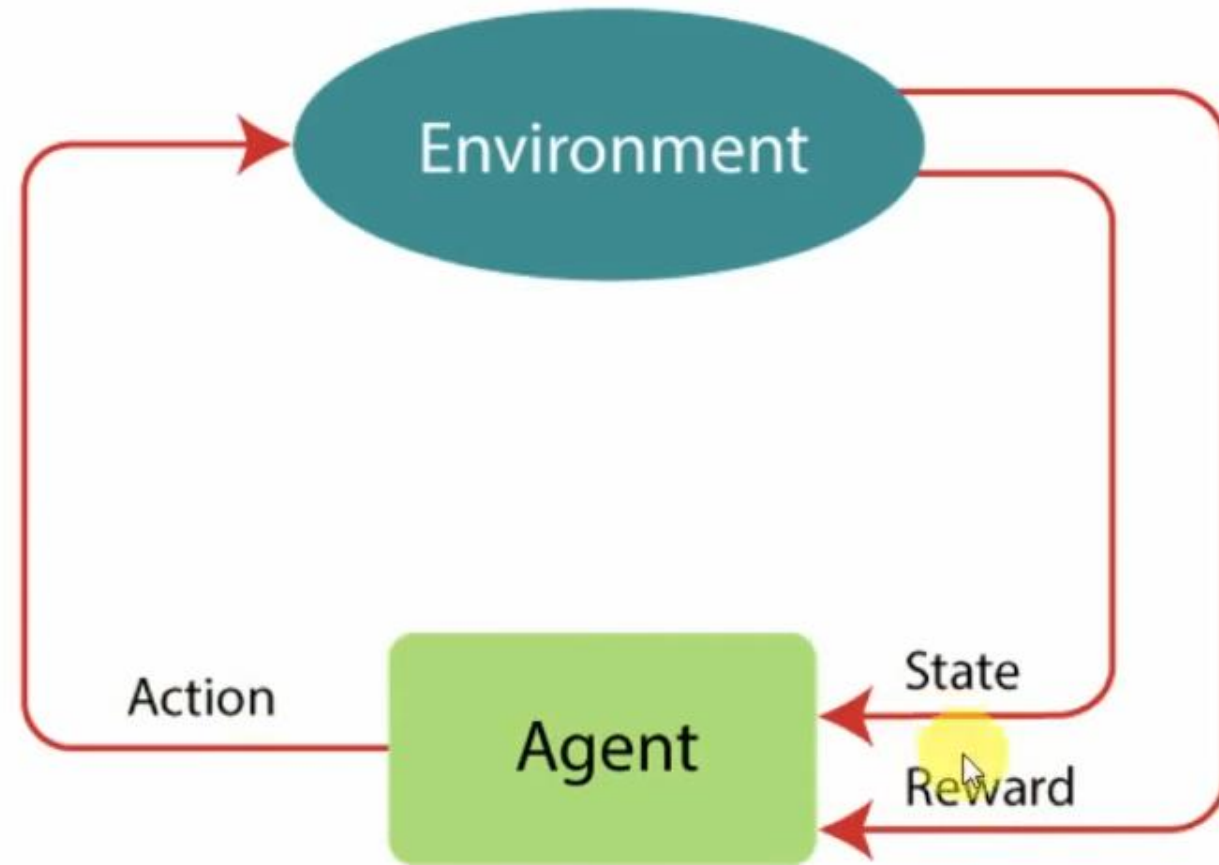
Dr. Maria Siddiqua
Assistant Professor
Department of AI & DS
FAST-NUCES Karachi
maria.siddiqua@nu.edu.pk

Course Contents

1. Introduction to AI
2. Agents and Environments
3. Problem Solving by Searching
4. Uninformed Search
5. Informed Search
6. Local Search
7. Constraint Satisfaction Problems
8. Adversarial Search
9. Knowledge Representation and Logical Agents
10. Uncertainty, Conditional Probability, Bayes Rule
11. Bayesian Networks, Naïve Bayes Classifier
12. Hidden Markov model, Markov Decision Process
- ➔ **13. Machine Learning (Supervised, Unsupervised, Reinforcement Learning)**

Introduction to Reinforcement Learning

- Reinforcement Learning is a feedback-based Machine learning Approach here an agent learns to which actions to perform by looking at the environment and the results of actions.
- For each correct action, the agent gets positive feedback, and for each incorrect action, the agent gets negative feedback or penalty.



Introduction to Reinforcement Learning

- The agent interacts with the environment and identifies the possible actions he can perform.
- The primary goal of an agent in reinforcement learning is to perform actions by looking at the environment and get the maximum positive rewards.
- In Reinforcement Learning, the agent learns automatically using feedbacks without any labeled data, unlike supervised learning.
- Since there is no labeled data, so the agent is bound to learn by its experience only.
- Reinforcement Learning is used to solve specific type of problem where decision making is sequential, and the goal is long-term, such as game-playing, robotics, etc.

Introduction to Reinforcement Learning

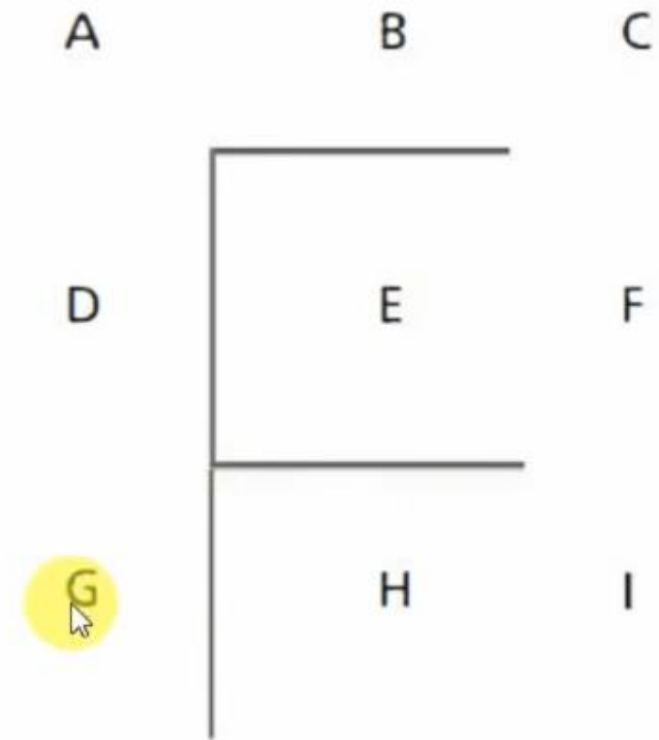
- There are two types of reinforcement learning - **positive and negative**.
- **Positive reinforcement learning** is a recurrence of behaviour due to positive rewards.
- Rewards increase strength and the frequency of a specific behaviour.
- This encourages to execute similar actions that yield maximum reward.
- Similarly, in **negative reinforcement learning**, negative rewards are used as a deterrent to weaken the behaviour and to avoid it.
- Rewards decreases strength and the frequency of a specific behaviour.

Introduction to Reinforcement Learning

- In a **maze game**, there may be a danger spot that may lead to loss.
- Negative rewards can be designed for such spots so that the agent does not visit that spot.
- Positive and negative rewards are simulated in reinforcement learning, say +10 for positive reward and -10 for some danger or negative reward.
- Reinforcement learning is an example of semi-supervised learning technique and is used to model sequential decision-making process.

Scope of Reinforcement Learning

- What are the situations where reinforcement learning can be used?
- Consider the following grid game shown in Figure , where a robot can move.
- Assume the starting node is E and goal node is G, the game is about finding the shortest path from starting to goal state.



Scope of Reinforcement Learning

- Reinforcement is not suitable for environments where complete information is available.
- For example, the problems like object detection, face recognition, fraud detection can be better solved using a classifier than by reinforcement learning.

Q-Learning Algorithm - Reinforcement learning

Q learning algorithm

For each s, a initialize the table entry $\hat{Q}(s, a)$ to zero

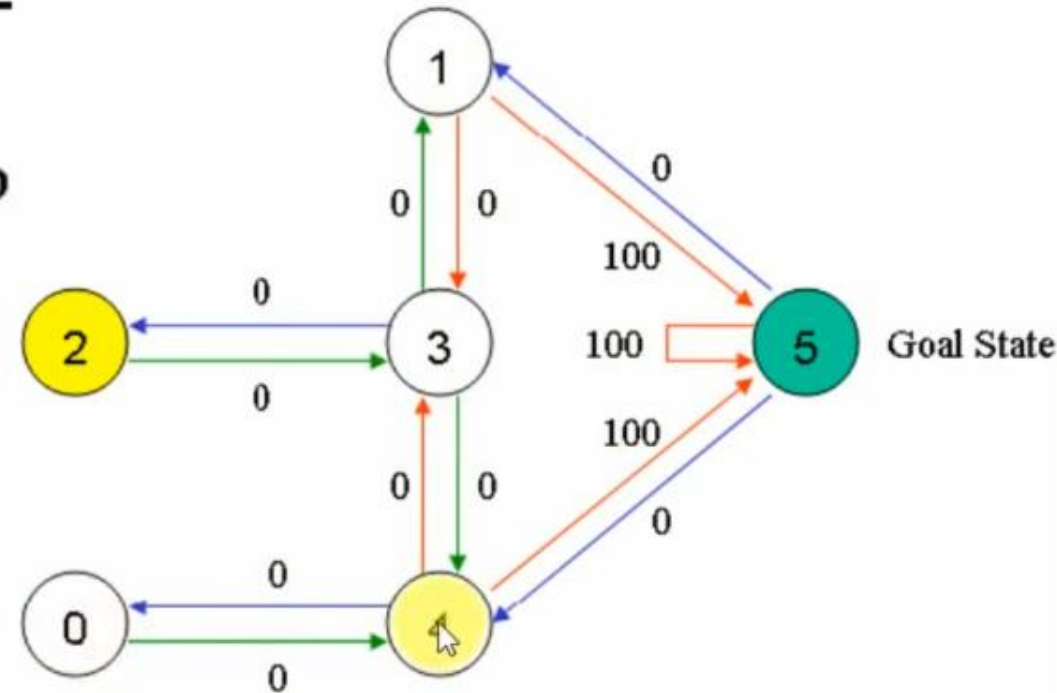
Observe the current state s

Do forever:

- Select an action a and execute it
- Receive immediate reward r
- Observe the new state s'
- Update the table entry for $\hat{Q}(s, a)$ as follows:

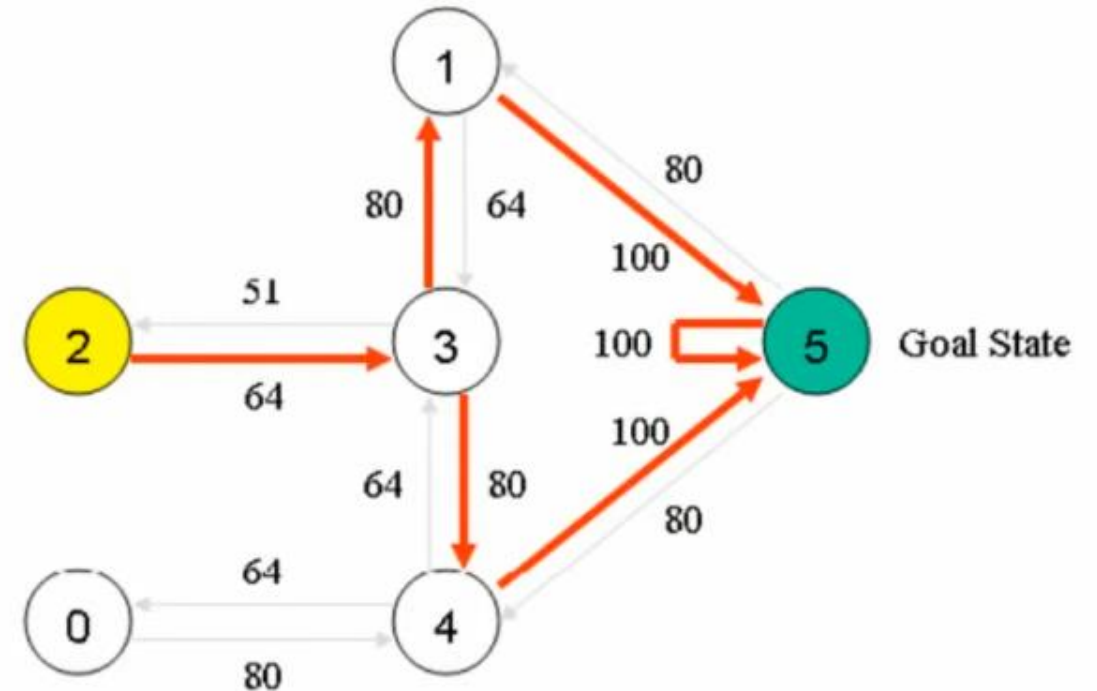
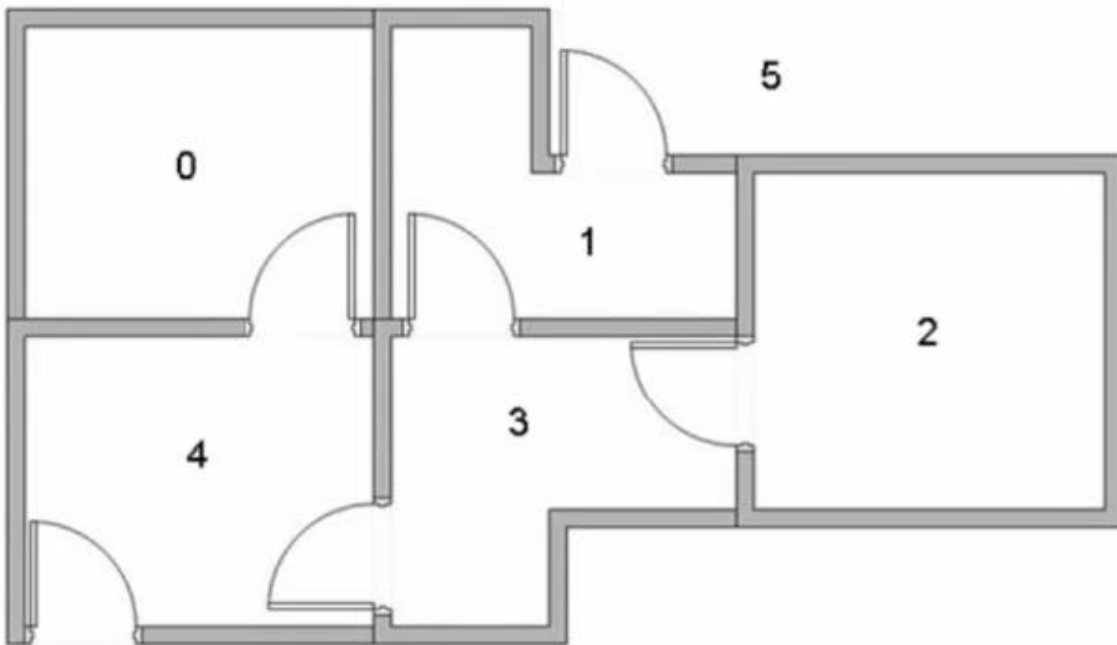
$$\hat{Q}(s, a) \leftarrow r + \gamma \max_{a'} \hat{Q}(s', a')$$

- $s \leftarrow s'$



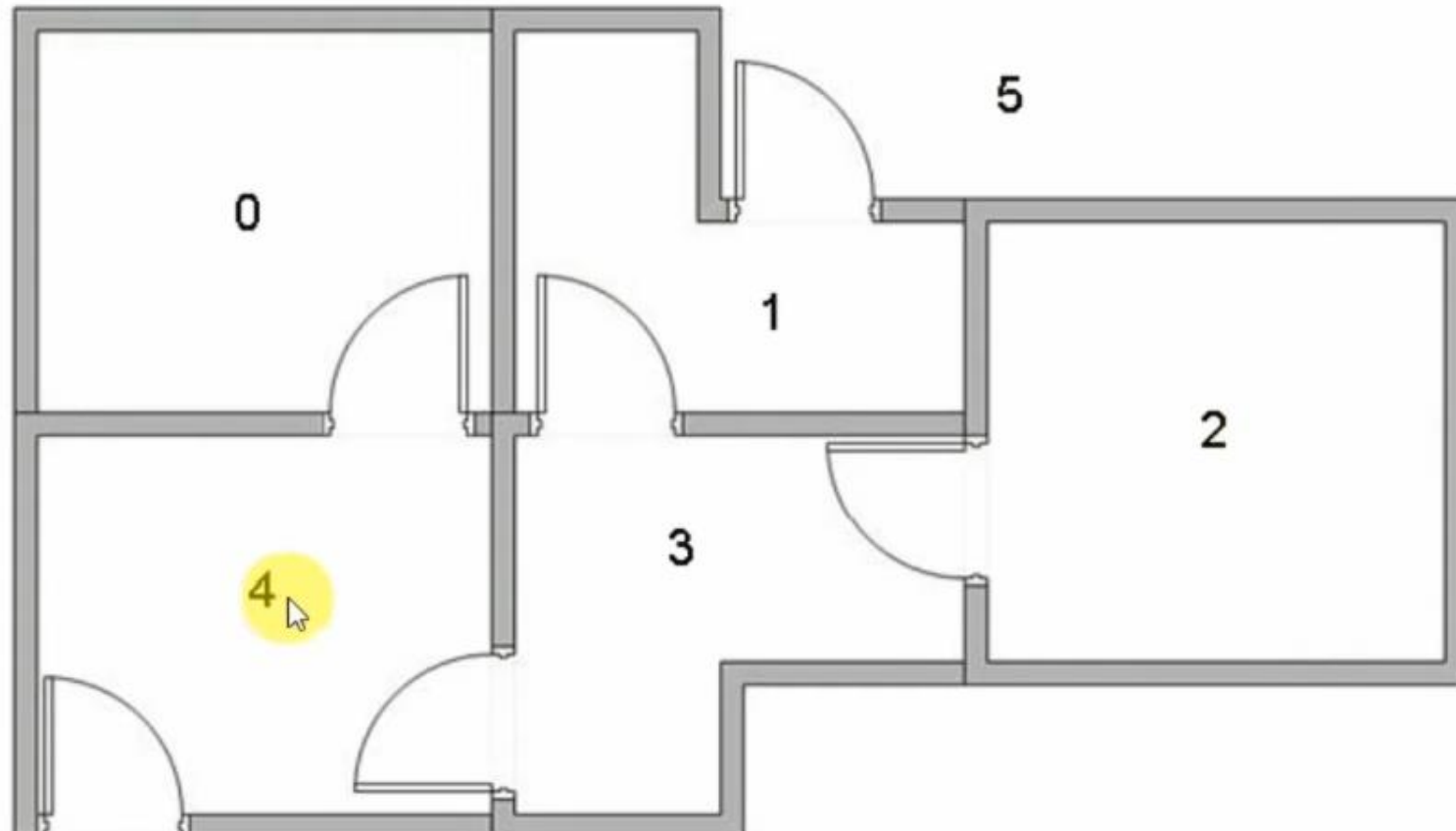
Q-Learning Algorithm

Solved Example - 1



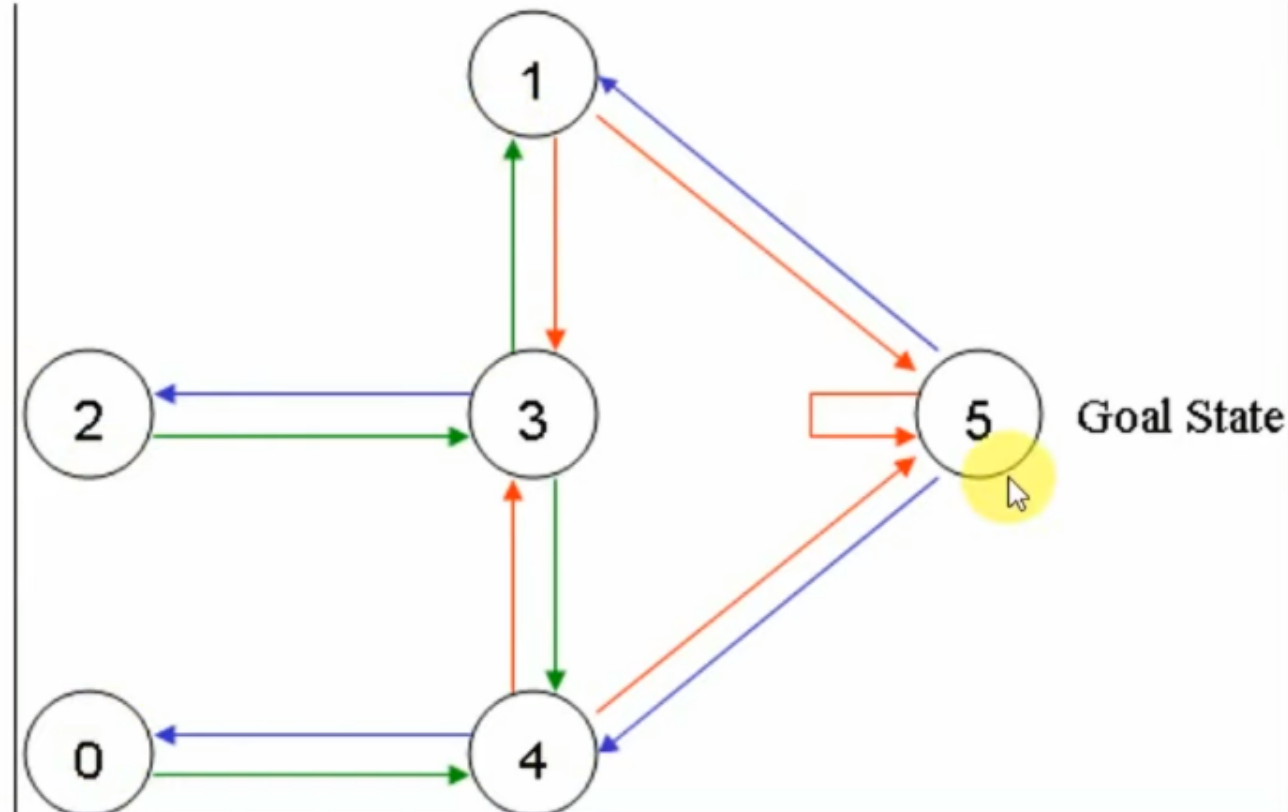
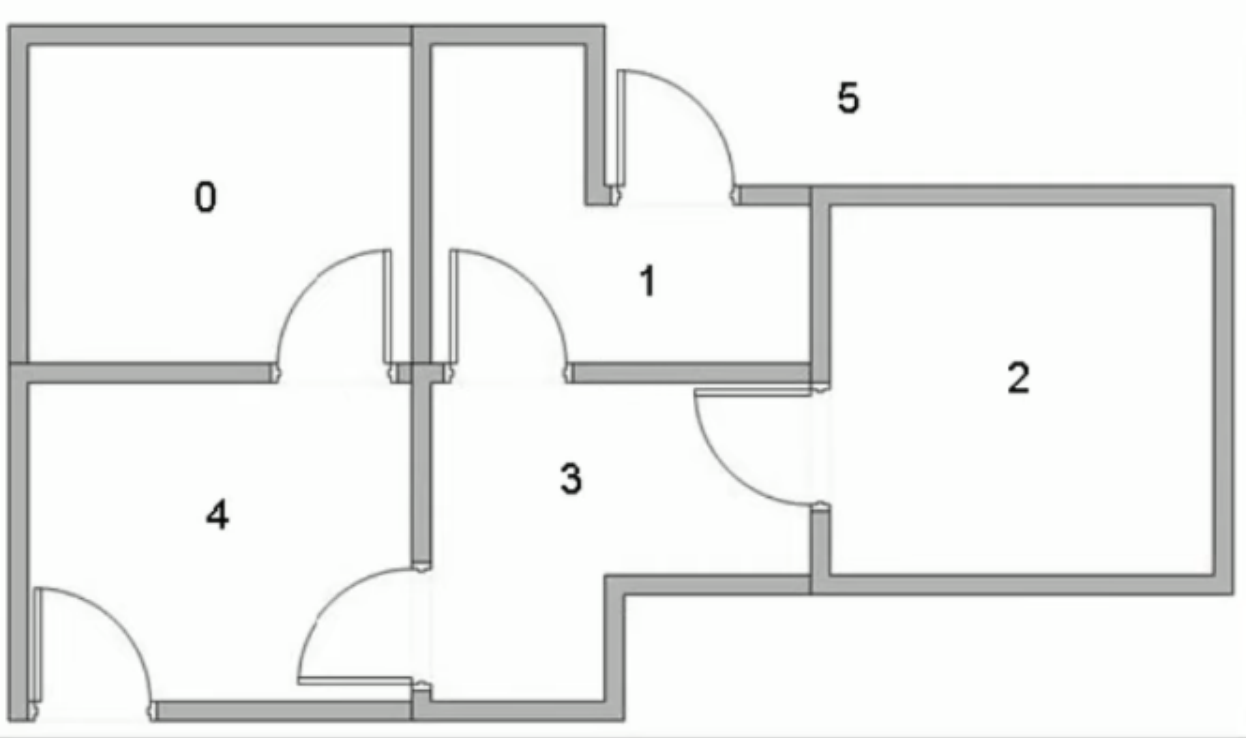
Q-Learning - Reinforcement Learning - Solved Example - 1

- Suppose we have 5 rooms in a building connected by doors as shown in the figure below. We'll number each room 0 through 4. The outside of the building can be thought of as one big room (5). Notice that doors 1 and 4 lead into the building from room 5 (outside).



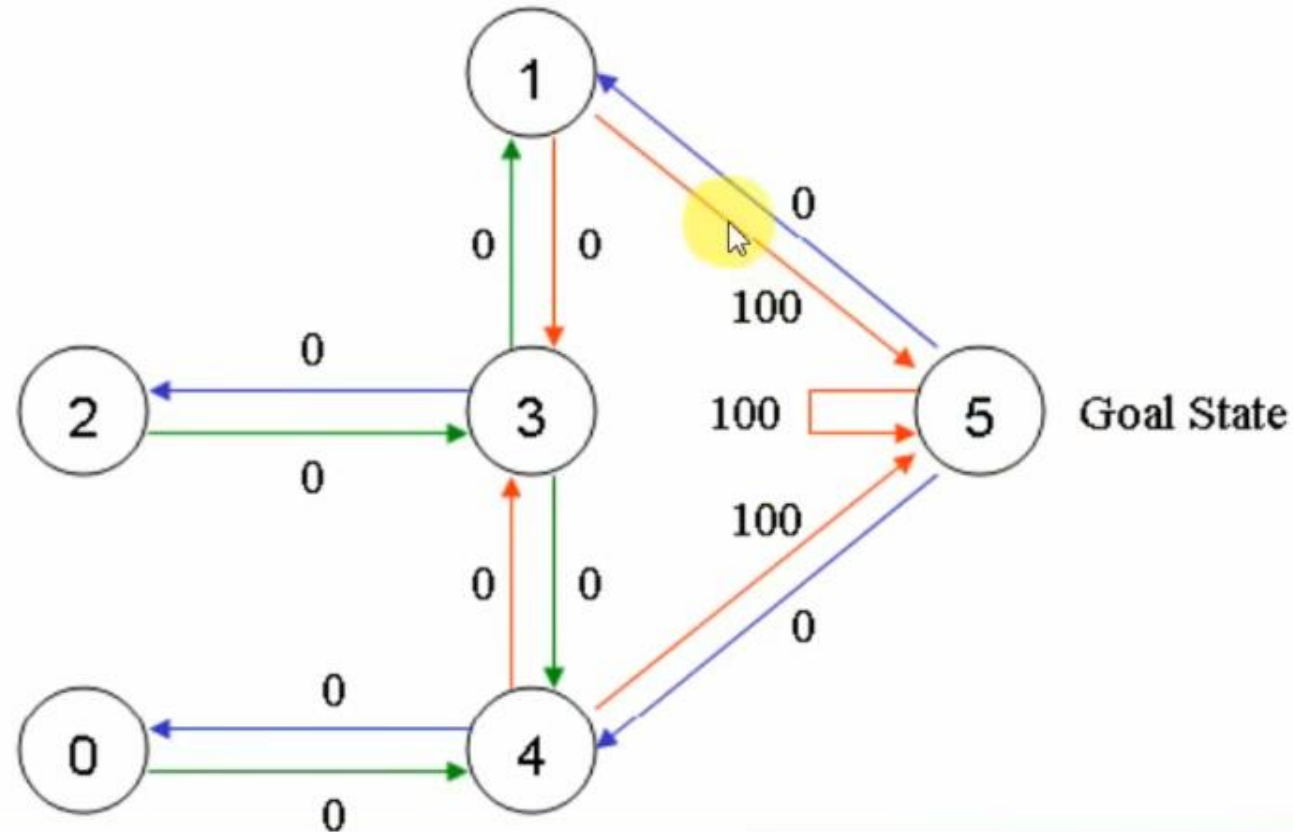
Q-Learning - Reinforcement Learning - Solved Example - 1

- We can represent the rooms on a graph, each room as a node, and each door as a link.



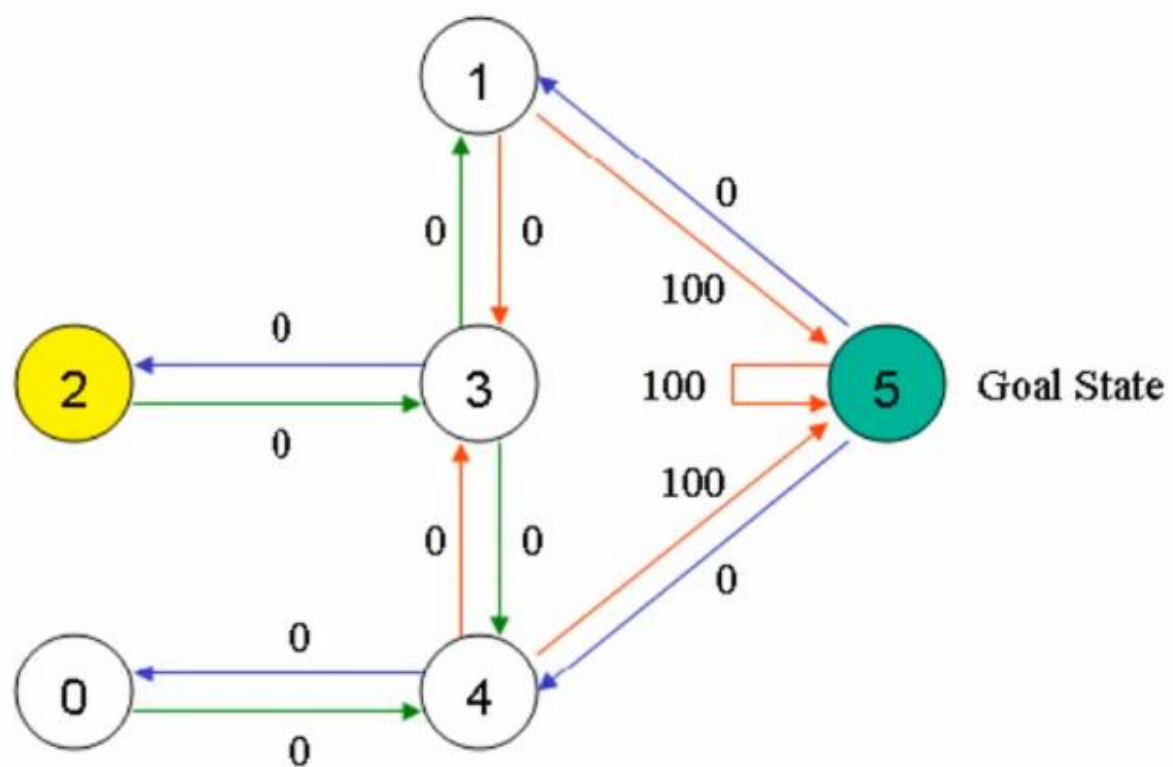
Q-Learning - Reinforcement Learning - Solved Example - 1

- The goal room is number 5
- The doors that lead immediately to the goal have an instant reward of 100. Other doors not directly connected to the target room have zero reward.
- Each arrow contains an instant reward value, as shown below:



Q-Learning - Reinforcement Learning - Solved Example - 1

- We can put the state diagram and the instant reward values into the following reward table, "matrix R". The -1's in the table represent null values (i.e.; where there isn't a link between nodes). For example, State 0 cannot go to State 1.



$R =$

State \ Action	0	1	2	3	4	5
0	-1	-1	-1	-1	0	-1
1	-1	-1	-1	0	-1	100
2	-1	-1	-1	0	-1	-1
3	-1	0	0	-1	0	-1
4	0	-1	-1	0	-1	100
5	-1	0	-1	-1	0	100

Q-Learning - Reinforcement Learning - Solved Example - 1

Q learning algorithm

For each s, a initialize the table entry $\hat{Q}(s, a)$ to zero.

Observe the current state s

Do forever:

- Select an action a and execute it
- Receive immediate reward r
- Observe the new state s'
- Update the table entry for $\hat{Q}(s, a)$ as follows:

$$\hat{Q}(s, a) \leftarrow r + \gamma \max_{a'} \hat{Q}(s', a')$$

- $s \leftarrow s'$

Q-Learning - Reinforcement Learning - Solved Example - 1

- Learning rate = 0.8 and the initial state as Room 1.
- Initialize matrix Q as a zero matrix:



Matrix R showing rewards for different states and actions.

State	0	1	2	3	4	5
0	-1	-1	-1	-1	0	-1
1	-1	-1	-1	0	-1	100
2	-1	-1	-1	0	-1	-1
3	-1	0	0	-1	0	-1
4	0	-1	-1	0	-1	100
5	-1	0	-1	-1	0	100

Q-Learning - Reinforcement Learning - Solved Example - 1

- **Gamma** = 0.8 and the initial state as Room 1.
- Initialize matrix Q as a zero matrix:

State	Action					
	0	1	2	3	4	5
0	-1	-1	-1	-1	0	-1
1	-1	-1	-1	0	-1	100
2	-1	-1	-1	0	-1	-1
3	-1	0	0	-1	0	-1
4	0	-1	-1	0	-1	100
5	-1	0	-1	-1	0	100

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

Q-Learning - Reinforcement Learning - Solved Example - 1

- Look at the second row (state 1) of matrix R.
- There are two possible actions for the current state 1: go to state 3, or go to state 5.
- By random selection, we select to go to 5 as our action.

		Action					
State		0	1	2	3	4	5
$R=$	0	-1	-1	-1	-1	0	-1
	1	-1	-1	-1	0	-1	100
	2	-1	-1	-1	0	-1	-1
	3	-1	0	0	-1	0	-1
	4	0	-1	-1	0	-1	100
	5	-1	0	-1	-1	0	100

Q-Learning - Reinforcement Learning - Solved Example - 1

- Now let's imagine what would happen if our agent were in state 5 (next state).
- Look at the sixth row of the reward matrix R (i.e. state 5).
- It has 3 possible actions: go to state 1, 4 or 5.
- $Q(\text{state}, \text{action}) = R(\text{state}, \text{action}) + \text{Gamma} * \text{Max}[Q(\text{next state}, \text{all actions})]$

State	Action					
	0	1	2	3	4	5
0	-1	-1	-1	-1	0	-1
1	-1	-1	-1	0	-1	100
2	-1	-1	-1	0	-1	-1
3	-1	0	0	-1	0	-1
4	0	-1	-1	0	-1	100
5	-1	0	-1	-1	0	100

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

Q-Learning - Reinforcement Learning - Solved Example - 1

- Now let's imagine what would happen if our agent were in state 5 (next state).
- Look at the sixth row of the reward matrix R (i.e. state 5).
- It has 3 possible actions: go to state 1, 4 or 5.
- $Q(\text{state}, \text{action}) = R(\text{state}, \text{action}) + \text{Gamma} * \text{Max}[Q(\text{next state}, \text{all actions})]$
- $Q(1, 5) = R(1, 5) + 0.8 * \text{Max}[Q(5, 1), Q(5, 4), Q(5, 5)] = 100 + 0.8 * 0 = 100$

State	Action					
	0	1	2	3	4	5
0	-1	-1	-1	-1	0	-1
1	-1	-1	-1	0	-1	100
2	-1	-1	-1	0	-1	-1
3	-1	0	0	-1	0	-1
4	0	-1	-1	0	-1	100
5	-1	0	-1	-1	0	100

Q	Action					
	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

Q-Learning - Reinforcement Learning - Solved Example - 1

- Now let's imagine what would happen if our agent were in state 5.
- Look at the sixth row of the reward matrix R (i.e. state 5).
- It has 3 possible actions: go to state 1, 4 or 5.
- $Q(\text{state}, \text{action}) = R(\text{state}, \text{action}) + \text{Gamma} * \text{Max}[Q(\text{next state}, \text{all actions})]$
- $Q(1, 5) = R(1, 5) + 0.8 * \text{Max}[Q(5, 1), Q(5, 4), Q(5, 5)] = 100 + 0.8 * 0 = 100$

State	Action					
	0	1	2	3	4	5
0	-1	-1	-1	-1	0	-1
1	-1	-1	-1	0	-1	100
2	-1	-1	-1	0	-1	-1
3	-1	0	0	-1	0	-1
4	0	-1	-1	0	-1	100
5	-1	0	-1	-1	0	100

	Action					
	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	0	0	100
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

Q-Learning - Reinforcement Learning - Solved Example - 1

- The next state, 5, now becomes the current state.
- Because 5 is the goal state, we've finished one episode.
- Our agent's brain now contains an updated matrix Q as:

$$Q = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 100 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Q-Learning - Reinforcement Learning - Solved Example - 1

- For the next episode, we randomly choose the initial state – say 3 (can go to 1, 2 & 4)

		Action					
State		0	1	2	3	4	5
$R=$	0	-1	-1	-1	-1	0	-1
	1	-1	-1	-1	0	-1	100
	2	-1	-1	-1	0	-1	-1
	3	-1	0	0	-1	0	-1
	4	0	-1	-1	0	-1	100
	5	-1	0	-1	-1	0	100

- Now we imagine that we are in state 1 (next state).

Q-Learning - Reinforcement Learning - Solved Example - 1

- Now we imagine that we are in state 1 (next state).
- Look at the second row of reward matrix R (i.e. state 1).
- It has 2 possible actions: go to state 3 or state 5.
- Then, we compute the Q value:
- $Q(\text{state}, \text{action}) = R(\text{state}, \text{action}) + \text{Gamma} * \text{Max}[Q(\text{next state}, \text{all actions})]$
- $Q(3, 1) = R(3, 1) + 0.8 * \text{Max}[Q(1, 3), Q(1, 5)] = 0 + 0.8 * \text{Max}(0, 100) = 80$

		Action					
State		0	1	2	3	4	5
0		-1	-1	-1	-1	0	-1
1		-1	-1	-1	0	-1	100
2		-1	-1	-1	0	-1	-1
3		-1	0	0	-1	0	-1
4		0	-1	-1	0	-1	100
5		-1	0	-1	-1	0	100

		0	1	2	3	4	5
0		0	0	0	0	0	0
1		0	0	0	0	0	100
2		0	0	0	0	0	0
3		0	0	0	0	0	0
4		0	0	0	0	0	0
5		0	0	0	0	0	0

Q-Learning - Reinforcement Learning - Solved Example - 1

- Now we imagine that we are in state 1 (next state).
- Look at the second row of reward matrix R (i.e. state 1).
- It has 2 possible actions: go to state 3 or state 5.
- Then, we compute the Q value:
- $Q(\text{state}, \text{action}) = R(\text{state}, \text{action}) + \text{Gamma} * \text{Max}[Q(\text{next state}, \text{all actions})]$
- $Q(3, 1) = R(3, 1) + 0.8 * \text{Max}[Q(1, 3), Q(1, 5)] = 0 + 0.8 * \text{Max}(0, 100) = 80$

State	Action					
	0	1	2	3	4	5
0	-1	-1	-1	-1	0	-1
1	-1	-1	-1	0	-1	100
2	-1	-1	-1	0	-1	-1
3	-1	0	0	-1	0	-1
4	0	-1	-1	0	-1	100
5	-1	0	-1	-1	0	100

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	0	0	0
2	0	0	0	0	0	100
3	0	80	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

Q-Learning - Reinforcement Learning - Solved Example - 1

- If our agent learns more through further episodes, it will finally reach convergence values in matrix Q like:

$$Q = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 80 & 0 \\ 0 & 0 & 0 & 64 & 0 & 100 \\ 0 & 0 & 0 & 64 & 0 & 0 \\ 0 & 80 & 51 & 0 & 80 & 0 \\ 64 & 0 & 0 & 64 & 0 & 100 \\ 0 & 80 & 0 & 0 & 80 & 100 \end{bmatrix} \end{matrix}$$

- Tracing the best sequences of states is as simple as following the links with the highest values at each state.

